

Question	Answer	Marks
11(a)	electrons (in gas atoms/molecules) interact with photons	B1
	photon energy causes electron to move to higher energy level/to be excited	B1
	photon energy = difference in energy of (electron) energy levels	B1
	when electrons de-excite, photons emitted in all directions (so dark line)	B1
11(b)(i)	photon energy $\propto 1 / \lambda$	C1
	energy = 1.68 eV	A1
	or	
	$E = hc / \lambda$ $E = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / (740 \times 10^{-9})$ $= 2.688 \times 10^{-19} \text{ J}$	(C1)
	energy = 1.68 eV	(A1)
11(b)(ii)	3.4 eV $\rightarrow$ 1.5 eV 3.4 eV $\rightarrow$ 0.85 eV 3.4 eV $\rightarrow$ 0.54 eV <i>all correct and none incorrect 2/2</i> <i>2 correct and 1 incorrect or only 2 correctly drawn 1/2</i>	B2

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12(c)	$x = 7$	M1